EXPERIMENT 4:

MOSFET CHARACTERISTICS

Date of Experiment : 10-08-2026 & 12-08-2026
Date of Submission : 20-08-2026

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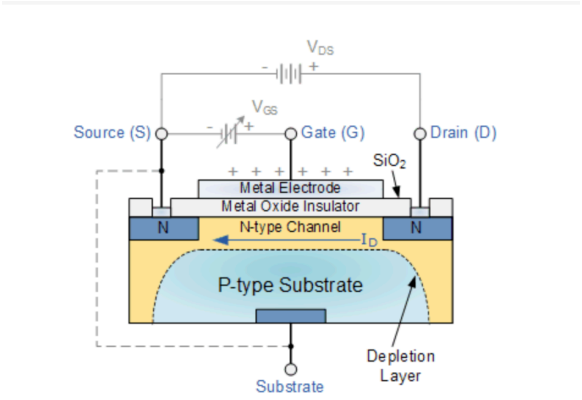
IMS24017

AIM:

To study drain and transfer characteristics of an n-channel Metal Oxide Semiconductor Field Effect Transistor (MOSFET) in common source configuration for different V_G values.

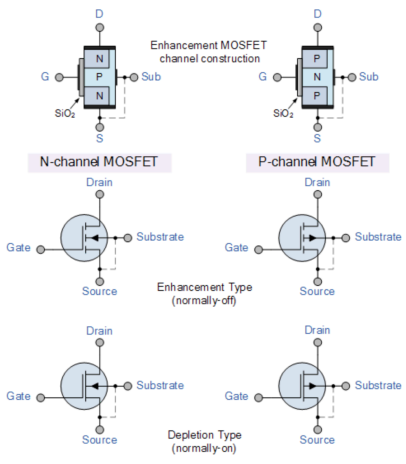
THEORY:



A Metal Oxide Semiconductor Field-Effect Transistor (MOSFET), also called an Insulated Gate Field-Effect Transistor (IGFET), is a voltage-controlled semiconductor device featuring three main terminals: Gate (G), Drain (D), and Source (S). Unlike Junction FETs, a MOSFET's gate electrode is completely isolated from the main conductive channel by an ultra-thin layer of insulating silicon dioxide (SiO_2). This insulation creates an extremely high input impedance (in the Mega-ohms range), meaning virtually no input current flows into the gate to control the conductive channel between the drain and source.

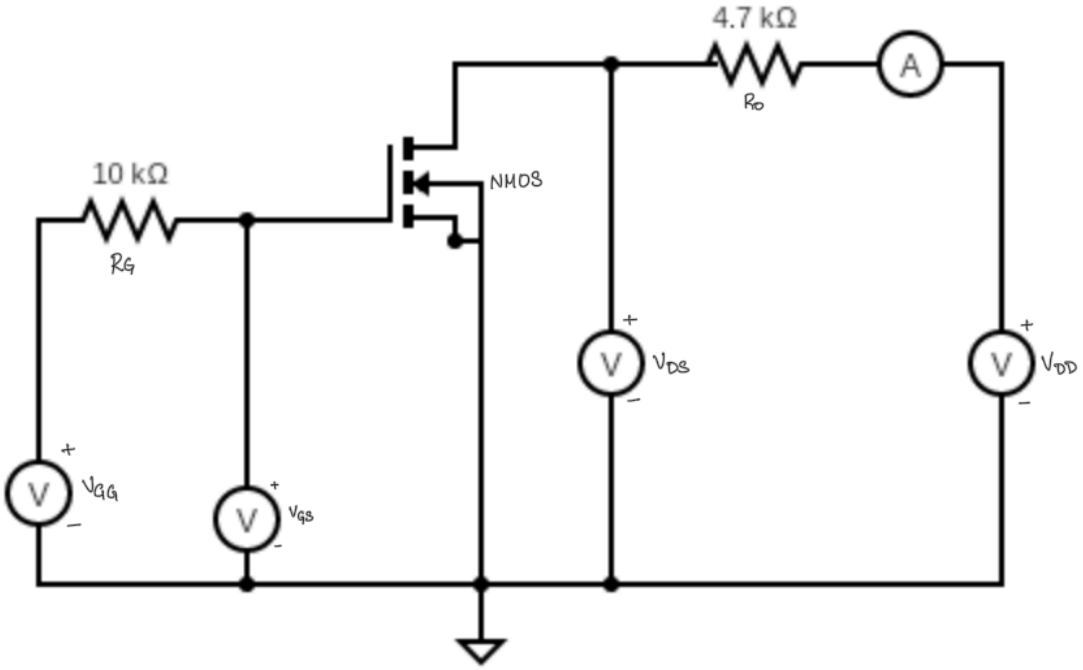
MOSFETs are categorised into N-channel (NMOS) or P-channel (PMOS) types and operate under two structural modes:

- Enhancement Mode (eMOSFET): Acts like a *normally-open* switch. It is "OFF" when the gate-source voltage (V_{GS}) is zero and requires a threshold voltage (V_{TH}) to create a conductive channel and turn's "ON".
- Depletion Mode (dMOSFET): Acts like a *normally-closed* switch. It naturally conducts at $V_{GS} = 0$ and requires a reverse-polarity gate bias to deplete free charge carriers and turn "OFF".



Depending on the applied biasing voltages, a MOSFET functions in three distinct regions: Cut-off ($V_{GS} < V_{TH}$, zero drain current; open switch), Linear/Ohmic (variable channel resistance; voltage-controlled resistor), and Saturation (maximum current flow; fully closed switch). Because of their low power consumption, high input resistance, and rapid switching capabilities, MOSFETs serve as core building blocks for both analog signal amplification (such as Class "A" amplifiers) and digital switching applications, including CMOS logic gates, microprocessors, and power control circuits.

CIRCUIT DIAGRAM:



PROCEDURE:

(i) Output/Drain Characteristics:

1. Wire up the MOSFET circuit according to the schematic. Connect an ammeter in series with the drain terminal and place voltmeters across the Gate-Source and Drain-Source terminals.
2. Adjust the gate power supply (V_{GG}) until V_{GS} reaches a chosen starting baseline. Hold this gate voltage constant for the entire first run.

3. Increase the main supply (V_{DD}) in fixed steps starting from zero to ramp up V_{DS}.
4. At every increment of V_{DS}, read the resulting drain current (I_D) from the ammeter and log both values in the observation table.
5. Turn V_{DS} back to zero, increase V_{GS} to the next step using V_{GG}, and repeat the V_{DS}sweep to generate a family of curves.
6. Plot V_{DS} on the X-axis against I_D on the Y-axis. Find dynamic drain resistance (r_d) by taking the inverse slope of the flat saturation region:

$$r_d = \left.\frac{\Delta V_{DS}}{\Delta I_D}\right|_{V_{GS}=\text{constant}}$$

(ii) <u>Transfer Characteristics:</u>

1. Keep the same basic circuit configuration, ensuring the meters are zeroed and leads are secured.
2. Adjust the main supply (V_{DD}) to set V_{DS} at a constant level high enough to keep the transistor operating in saturation.
3. Starting from zero volts, slowly increase V_{GS} in small, precise steps using the gate power supply (V_{GG}).
4. Observe the ammeter closely. Keep logging $I_D = 0$ until current begins to flow, then record I_D at each V_{GS} step as the channel conducts.
5. Plot V_{GS} on the X-axis against I_D on the Y-axis to form the characteristic transconductance curve.
6. Determine the threshold voltage (V_{TH}) at the X-axis intercept where I_D begins flowing, and calculate transconductance (g_m) from the slope:

$$g_m = \left.\frac{\Delta I_D}{\Delta V_{GS}}\right|_{V_{DS}=\text{constant}}$$

OBSERVATIONS:

(i) Output Characteristics:

Table 1:

$V_{DS}(V)$	$V_{GS} = 3.5V$ $I_D(mA)$	$V_{GS} = 3.6V$ $I_D(mA)$	$V_{GS} = 3.7V$ $I_D(mA)$
0	0	0	0
0.2	0.48	0.55	0.49
0.5	0.89	1.03	1.13
1	1.07	2.03	2.36
2	1.12	2.49	4.38
3	1.17	2.58	6.09
4	1.22	2.62	6.86
5	1.27	2.69	7.01
6	1.33	2.74	7.14
7	1.36	2.81	7.29
8	1.41	2.89	7.43
9	1.46	2.96	7.55
10	1.51	3.01	7.68

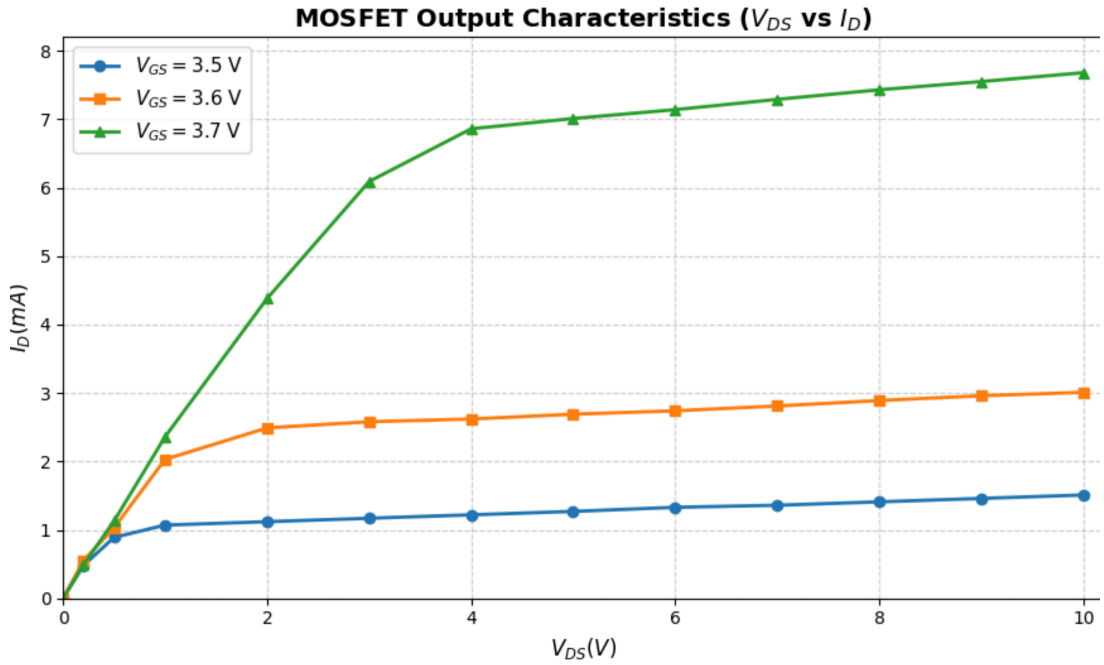
(ii) Transfer Characteristics:

Table 2:

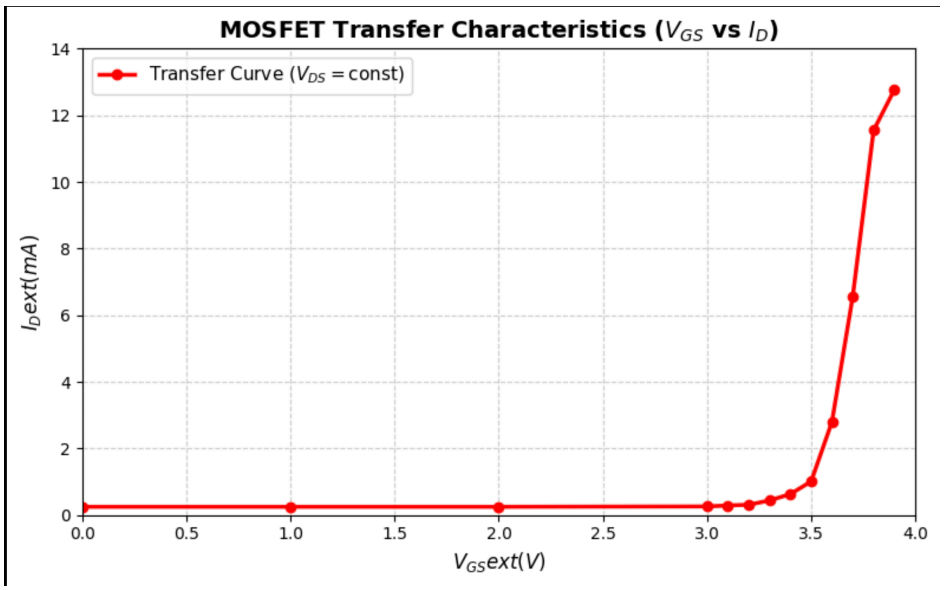
$V_{GS}(V)$	$I_D(mA)$
0	0.25
1	0.25
2	0.25
3	0.26
3.1	0.29
3.2	0.31

$V_{GS}(V)$	$I_D(mA)$
3.3	0.44
3.4	0.63
3.5	1.01
3.6	2.80
3.7	6.57
3.8	11.55
3.9	12.77

Graph 1:



Graph 2:



CALCULATIONS:

Dynamic Drain Resistance:

$$r_d = \left.\frac{\Delta V_{DS}}{\Delta I_D}\right|_{V_{GS}=\text{constant}}$$

$$= \frac{V_{DS2} - V_{DS1}}{I_{D2} - I_{D1}}$$

$$= \frac{10.0\ \text{V} - 4.0\ \text{V}}{3.01\ \text{mA} - 2.62\ \text{mA}}$$

$$= \frac{6.0\ \text{V}}{0.39\ \text{mA}}$$

$$\approx 15.38\ \text{k}\Omega \quad (15{,}384.6\ \Omega)$$

Transconductance:

$$g_m = \left.\frac{\Delta I_D}{\Delta V_{GS}}\right|_{V_{DS}=\text{constant}}$$

$$= \frac{I_{D2} - I_{D1}}{V_{GS2} - V_{GS1}}$$

$$= \frac{11.55\ \text{mA} - 1.01\ \text{mA}}{3.8\ \text{V} - 3.5\ \text{V}}$$

$$= \frac{10.54\ \text{mA}}{0.3\ \text{V}}$$

$$= 35.13\ \text{mS} \quad (35.13 \times 10^{-3}\ \text{S})$$

RESULT:

I. The threshold voltage $(V_{TH}) \approx 3.50$ V
II. Transconductance $(g_m) \approx 35.13$ mS
III. Drain Resistance $(r_d) \approx 15.38$ kΩ

DISCUSSION AND REMARKS:

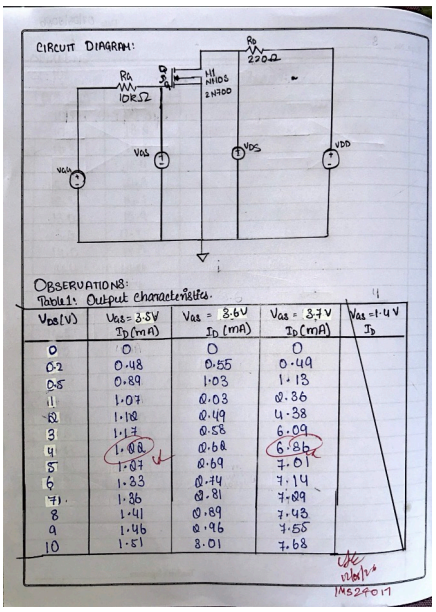
The experimental output characteristics clearly demonstrate the core operational regions of an N-channel enhancement-mode MOSFET. At low drain-to-source voltages, the device operates in the ohmic region where the conductive channel behaves as a voltage-controlled resistor, causing the drain current to rise sharply with increasing drain voltage. As the drain voltage increases further, the potential gradient across the gate oxide causes channel pinch-off near the drain terminal, forcing the transistor into saturation. In this saturation region, the drain current flattens out and becomes nearly independent of further increases in drain voltage, yielding a high dynamic drain resistance.

The transfer characteristics curve illustrates gate control over channel conductivity when the drain voltage is held constant in the saturation region. Below the threshold voltage, the channel remains non-conductive with only a negligible baseline leakage current. Once the gate-to-source voltage exceeds the threshold level, strong inversion occurs within the semiconductor substrate, causing the drain current to increase rapidly in a parabolic fashion. Evaluating the slope of this curve in its steep region highlights the forward transconductance of the device, confirming strong gate-voltage modulation over charge carrier transport.

Comparing data across continuous test runs reveals minor physical discrepancies typical of laboratory bench measurements. Specifically, a slight reduction in peak drain current between successive measurement sets at equivalent gate voltages is primarily driven by junction self-heating during continuous DC sweeps. As temperature rises within the device, thermal agitation in the silicon lattice reduces electron mobility, slightly lowering overall channel conductance. Additionally, any small non-zero current observed prior to threshold onset stems from minor meter zero-offset calibration or sub threshold leakage near the oxide boundary.

Overall, the experimental results closely align with theoretical MOSFET behaviour and validate the efficacy of the insulated-gate structure. The silicon dioxide insulating layer maintains an extremely high input impedance, drawing virtually zero gate current during static control. Because channel conductance can be precisely modulated using small gate voltage variations, the device demonstrates its dual functionality as an efficient voltage-controlled switch across cut-off and ohmic states, and as a linear amplifier when biased within the saturation region.

LAB RECORD:



CIRCUIT DIAGRAM:

OBSERVATIONS:

Table 1: Output characteristics.

V_{DS} (V)	V_{GS} = 3.5V I_D (mA)	V_{GS} = 3.6V I_D (mA)	V_{GS} = 3.7V I_D (mA)	V_{GS} = 1.4V I_D
0	0	0	0	
0.2	0.48	0.55	0.49	
0.5	0.89	1.03	1.13	
1	1.07	0.03	0.36	
2	1.18	0.49	4.38	
3	1.17	0.58	6.09	
4	1.08	0.68	6.86	
5	1.07	0.69	7.01	
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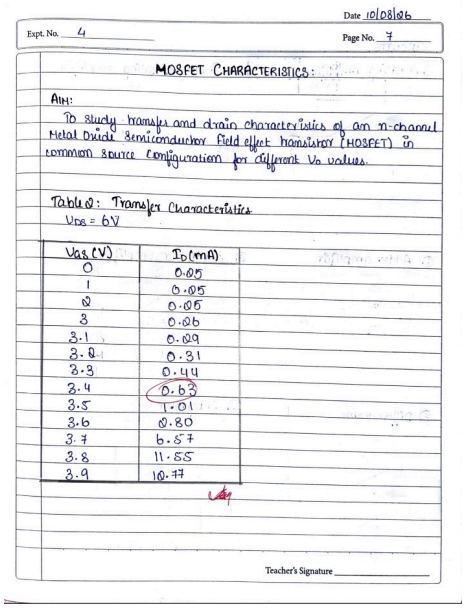

Date 10/08/06

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MOSFET CHARACTERISTICS:

Aim:

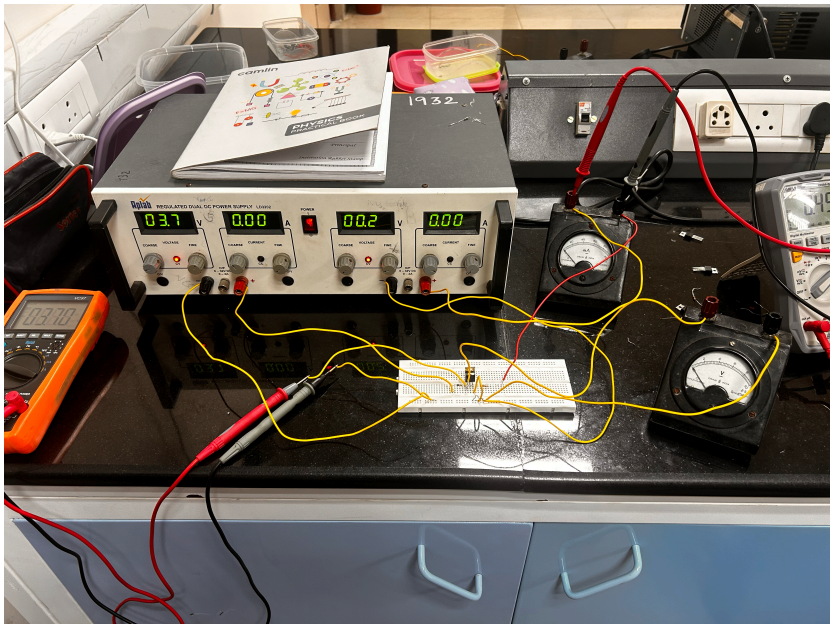
To study transfer and drain characteristics of an n-channel Metal Oxide Semiconductor Field effect transistor (MOSFET) in common source configuration for different V_{DS} values.

Table 2: Transfer Characteristics

V_{DS} = 6V

V_{GS} (V)	I_D (mA)
0	0.05
1	0.05
2	0.05
3	0.06
3.1	0.09
3.2	0.31
3.3	0.44
3.4	0.63
3.5	1.01
3.6	2.80
3.7	6.57
3.8	11.55
3.9	10.77

Teacher's Signature



REFERENCES:

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- Lab Manual